HOMEWORK 6
INTRODUCTION TO COMPLEX ANALYSIS
(MATH 4230-40), SPRING 2022

SECTION 2.3 - A CLOSER LOOK AT CAUCHY'S THEOREM

1. Prove that $\mathbb{C} \setminus \left\{0\right\}$ is not simply connected.
2. Show that every disk is convex.
3. Evaluate the following integrals without performing explicit computation:
   (a) $\displaystyle\int_\gamma \frac{dz}{z}$, where $\gamma(t) = \cos t + 2i \sin t$, $0 \le t \le 2\pi$.
   (b) $\displaystyle\int_\gamma \frac{dz}{z^2}$, where $\gamma(t) = \cos t + 2i \sin t$, $0 \le t \le 2\pi$.
   (c) $\displaystyle\int_\gamma \frac{e^z dz}{z}$, where $\gamma(t) = 2 + e^{it}$, $0 \le t \le 2\pi$.